

SOFTWARE TEST PLAN (STP)

Online Quiz System

Console-based Multiple-Choice Quiz Application (C / C++)

Project	Online Quiz System
Version	1.0
Date	16-09-2026
Status	Test Plan - Submission Copy

Prepared for Software Engineering Mini-Project - Test Plan Deliverable

1. Introduction

Purpose: This document defines the test plan for the Online Quiz System v1.0. It specifies what will be tested, the test approach, environment, schedule, responsibilities, risks, traceability, metrics, and acceptance conditions.

Scope: Testing covers application launch, player-name input, question loading, question display, answer validation, scoring, progress, final result, replay, clean exit, performance, reliability, usability, portability, and basic security/input-safety checks. Graphical/web interfaces, networking, cloud services, databases, and multi-user operation are outside the scope.

References: Online Quiz System project requirements; Software Test Plan template supplied for the mini-project; approved project source/design artefacts.

Definitions: OQS - Online Quiz System; MCQ - Multiple-Choice Question; CLI - Command-Line Interface; STP - Software Test Plan; FR - Functional Requirement; NFR - Non-Functional Requirement; RTM - Requirements Traceability Matrix; UAT - User Acceptance Testing.

2. Test Items

- Application launch and main console interface
- Player-name input and validation
- Question-bank loading and validation
- Quiz engine and question presentation
- Answer capture and invalid-input handling
- Answer evaluation, scoring, progress, and percentage calculation
- Final result, replay, and exit control
- Non-functional qualities: performance, reliability, usability, and portability
- Security and input-safety checks

3. Features to be Tested

Requirement / Feature ID	Feature to be Tested	Test Case
OQS-F-001	Application starts and displays the quiz/main menu	TC-INIT-01
OQS-F-002	Player name is captured and validated	TC-INIT-02
OQS-F-004	Question bank is loaded successfully	TC-QB-01
OQS-F-006	One question is shown at a time with a question number	TC-QB-01
OQS-F-008	Valid answer option is accepted	TC-ANS-01
OQS-F-009	Invalid answer is rejected without advancing	TC-ANS-02
OQS-F-012	Score increases only for a correct answer	TC-SCORE-01
OQS-F-014	Quiz progress is displayed correctly	TC-SCORE-01
OQS-F-016/017	Final score and percentage are calculated correctly	TC-RES-01
OQS-F-018	Final result summary is displayed	TC-RES-01
OQS-F-019/020	Replay starts a fresh attempt and the program can exit cleanly	TC-CTRL-01
OQS-NF-001	Application starts within the expected response time	TC-NF-01
OQS-NF-003	Invalid inputs do not crash or corrupt the session	TC-SEC-01

4. Features Not to be Tested

- Graphical, web, or mobile user interfaces
- Network communication, client-server behavior, cloud services, or remote APIs
- Database-server behavior and distributed/multi-user operation
- Operating-system, compiler, terminal, or standard-library internals
- Any functionality not part of the base Online Quiz System such as online accounts, payments, or multiplayer mode

5. Test Approach / Strategy

Levels: Unit testing will check individual functions such as input validation and scoring; integration testing will verify interaction between question loading, quiz flow, input, and scoring; system testing will cover complete quiz execution; acceptance testing will confirm the application is ready for submission.

Types: Functional, regression, performance, reliability/robustness, usability, portability/build, and security/input-safety testing will be performed.

Entry Criteria: A compilable and runnable build is available, test data is prepared, the console environment and compiler are ready, and planned test cases have expected results.

Exit Criteria: All planned test cases are executed; all core/high-priority quiz functions pass; no critical defect remains open; failed cases are retested after fixes; and the final test summary shows the system is ready for evaluation.

5.1 Security Validation

- Provide empty, malformed, out-of-range, and unusually long input and verify that it is rejected safely.
- Check first/last valid answer positions and invalid indexes to ensure no out-of-range access occurs.
- Use malformed question records and verify they are rejected or reported instead of being used.
- Enter text resembling shell commands or code fragments and verify it is handled only as ordinary input.
- Verify the score remains between 0 and the number of questions and is reset when replay starts.
- Repeat boundary and negative tests after changes to input, scoring, question parsing, or replay logic.

6. Test Environment

Item	Planned Environment
Hardware	Desktop/laptop computer, keyboard, and monitor
Operating System	Windows 10/11 or Linux terminal environment
Software Under Test	Online Quiz System v1.0 compiled from project source
Compiler / Runtime	Standards-compliant C11 or C++17 compiler, e.g., GCC/MinGW or equivalent
Tools	Terminal/command prompt, compiler warnings, debugger if required, simple timing tool, and a defect log
Test Data	Valid question bank; malformed records; valid/invalid names; correct/incorrect/boundary answer inputs; replay and exit scenarios
Network	Not required

7. Test Schedule

Milestone	Planned Date
Test plan and test-case design	16-Sep-2026
Test environment and data setup	17-Sep-2026
Unit and integration testing	18-Sep-2026
System and functional testing	19-Sep-2026 to 20-Sep-2026
NFR and security validation	21-Sep-2026
Defect fixing and regression	22-Sep-2026
UAT and final test summary	23-Sep-2026

8. Test Deliverables

- Software Test Plan (this document)
- Test Case Specification
- Test data / question-bank variants
- Test execution evidence or screenshots
- Defect and retest records, if defects are found
- Requirements Traceability Matrix (RTM)
- Final Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	_____	Prepare the plan, coordinate testing, monitor progress, and confirm exit criteria.
Test Engineer	_____	Design and execute test cases, record evidence, and log/retest defects.
Developer	_____	Provide testable builds, fix defects, and support regression testing.
Faculty / Evaluator	_____	Review evidence and validate acceptance readiness.

10. Risks and Mitigation

Risk	Mitigation
Delay in a stable build	Use an early smoke build and prioritize blocking defects.
Malformed or insufficient question data	Prepare a known-valid bank plus intentionally invalid records before execution.
Compiler/platform differences	Compile early on the target lab machine and use standard C/C++ features.
Input-handling defects	Run boundary, negative, and repeated invalid-input tests before acceptance testing.
Scoring or replay defects	Use deterministic answers with known expected scores and repeat replay tests after fixes.
Limited time before submission	Execute core/high-priority tests first and maintain a simple defect/retest status.

11. Assumptions & Dependencies

- A compilable C/C++ project and the required compiler/toolchain are available before formal testing.
- A valid question bank with enough questions for end-to-end testing is available.
- The terminal accepts keyboard input and displays text correctly.
- Expected answers used for testing are known and controlled.
- No network, cloud service, or external database is required for the base project.

12. Suspension & Resumption Criteria

Suspend testing if the application cannot compile or launch, a blocking defect prevents a large portion of planned cases, required test data is unavailable, the test environment is unavailable for an extended period, or repeated crashes make results unreliable.

Resume testing after the blocking issue is fixed, the environment/test data is restored, and a smoke test confirms that launch, question loading, answer input, scoring, and exit are usable again.

13. Test Case Management & Traceability

Each test case records the feature/requirement, preconditions, test data, steps, expected result, actual result, status, and evidence/defect reference. The following mapping provides traceability between the planned features and the selected test cases.

Coverage Area	Requirement / Feature	Test Case(s)
Initialization / player input	OQS-F-001, OQS-F-002	TC-INIT-01, TC-INIT-02
Question loading / presentation	OQS-F-004, OQS-F-006	TC-QB-01
Answer input / validation	OQS-F-008, OQS-F-009	TC-ANS-01, TC-ANS-02
Scoring / progress	OQS-F-012, OQS-F-014	TC-SCORE-01
Final result / percentage	OQS-F-016 to OQS-F-018	TC-RES-01
Replay / exit	OQS-F-019, OQS-F-020	TC-CTRL-01
Non-functional / security	OQS-NF-001, OQS-NF-003	TC-NF-01, TC-SEC-01

13.1 Detailed Test Cases

TC-INIT-01 - Application Launch

Requirement / Feature	OQS-F-001
Preconditions	Executable has been built successfully.
Test Data	No special input.
Steps	1. Start the program. 2. Observe the first screen/menu.
Expected Result	Program launches without error and displays the Online Quiz System start/main menu.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-INIT-02 - Player Name Validation

Requirement / Feature	OQS-F-002
Preconditions	Application is running and player-name prompt is displayed.
Test Data	Valid name: Jishnu; invalid input: empty entry.
Steps	1. Enter an empty name. 2. Verify a re-prompt. 3. Enter a valid name.
Expected Result	Empty input is rejected; valid non-empty name is accepted and the quiz continues.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-QB-01 - Question Loading and Presentation

Requirement / Feature	OQS-F-004, OQS-F-006
Preconditions	A valid question bank is available.
Test Data	Known question bank containing multiple MCQs.
Steps	1. Start a quiz. 2. Observe the first and next questions.
Expected Result	Questions load successfully and only one current question is presented at a time with a visible question number and options.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-ANS-01 - Accept Valid Answer

Requirement / Feature	OQS-F-008
Preconditions	A question with valid answer options is displayed.
Test Data	Any displayed valid option, e.g., A.
Steps	1. Enter a valid displayed option. 2. Submit the answer.
Expected Result	The answer is accepted and recorded; evaluation/progress continues.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-ANS-02 - Reject Invalid Answer

Requirement / Feature	OQS-F-009
Preconditions	A question is displayed.
Test Data	Invalid input such as Z or 9 when only A-D / 1-4 are allowed.
Steps	1. Enter invalid input. 2. Observe the program state. 3. Enter a valid answer.
Expected Result	Invalid input is rejected with a clear message; the question does not advance until a valid answer is entered.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-SCORE-01 - Scoring and Progress

Requirement / Feature	OQS-F-012, OQS-F-014
Preconditions	A quiz with a known answer key is running.
Test Data	One known correct answer followed by one known incorrect answer.
Steps	1. Submit a correct answer. 2. Observe score/progress. 3. Submit an incorrect answer. 4. Observe score/progress again.
Expected Result	Score increases only for the correct answer; progress/question number advances correctly after each accepted answer.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-RES-01 - Final Score and Percentage

Requirement / Feature	OQS-F-016, OQS-F-017, OQS-F-018
Preconditions	Complete quiz with a known number of correct answers.
Test Data	Example: 4 correct answers out of 5 questions.
Steps	1. Complete the quiz with the planned answer set. 2. Observe the final result.
Expected Result	Final score is 4/5 and percentage is 80%; player/result summary fields are shown correctly.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-CTRL-01 - Replay and Clean Exit

Requirement / Feature	OQS-F-019, OQS-F-020
Preconditions	A quiz has been completed and replay/exit controls are available.
Test Data	Choose Replay / Yes, then later choose Exit.
Steps	1. Finish one attempt. 2. Select replay and verify a fresh attempt. 3. Return to the exit option. 4. Choose Exit.
Expected Result	Replay starts a fresh attempt with score/state reset; choosing Exit then terminates the program normally without a crash or hanging prompt.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-NF-01 - Startup Performance

Requirement / Feature	OQS-NF-001
Preconditions	Application and test question bank are ready on the target lab system.
Test Data	Normal valid question bank.
Steps	1. Measure time from program start to first usable menu/question. 2. Repeat several times.
Expected Result	Startup/first usable response remains within the planned target of approximately 2 seconds under normal local execution.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

TC-SEC-01 - Invalid / Long Input Robustness

Requirement / Feature	Security/Input Safety and OQS-NF-003
Preconditions	Application is running at an input prompt.
Test Data	Empty input, out-of-range input, malformed text, and a long input string.
Steps	1. Enter each invalid/boundary input. 2. Observe application behavior and score/session state. 3. Continue with a valid input.
Expected Result	Inputs are rejected or handled safely; the application does not crash, advance incorrectly, or corrupt the score/session state.
Actual Result	To be recorded during execution
Status	Not Run
Evidence / Defect ID	To be attached if applicable

14. Test Metrics & Reporting

- Number and percentage of planned test cases executed
- Number and percentage passed, failed, blocked, and not run
- Requirement/feature coverage percentage
- Defect count by severity and status
- Regression/retest pass rate after fixes
- Measured startup/response results for applicable NFR tests

Reports: A simple execution status/defect record will be maintained during testing, followed by a final Test Summary Report containing coverage, test results, open issues, and exit-criteria status.

15. Approvals

Role	Name	Signature	Date
QA Lead			
Developer / Dev Lead			
Faculty / Evaluator			